

Brugada Syndrome 7 (BrS7): A Comprehensive Disease Characterization Report

Disease: Brugada syndrome 7 · **Gene:** *SCN3B* (Navβ3) · **OMIM:** #613120 · **Category:** Genetic (cardiac channelopathy) **Evidence status:** *SCN3B* has "limited" gene-disease validity for Brugada syndrome (*SCN5A* is the sole "definitive" gene)

Autonomous literature-based investigation. Evidence types are flagged as [human clinical], [in vitro], [model organism], [computational], or [database]. Where primary data specific to the SCN3B/BrS7 subtype are lacking, general Brugada-syndrome (BrS) evidence is used and labeled as such.

Summary

Brugada syndrome 7 (BrS7; OMIM #613120) is a rare, autosomal-dominant cardiac sodium channelopathy attributed to loss-of-function variants in *SCN3B*, the gene encoding the auxiliary β3 subunit (Navβ3) of the cardiac voltage-gated sodium channel. Mutant β3 fails to properly chaperone and modulate the pore-forming Nav1.5 (*SCN5A*) α-subunit, reducing its cell-surface expression and cardiac sodium current (I_{Na}) [in vitro,

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The reduced inward sodium current unmasks transient-outward-current (I_{to})-mediated repolarization heterogeneity in the right ventricular outflow tract (RVOT), which generates the diagnostic type-1 (coved) Brugada ECG pattern and creates a phase-2-reentry substrate for polymorphic ventricular tachycardia (VT), ventricular fibrillation (VF), and sudden cardiac death (SCD) [in vitro,

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Clinically, BrS7 is indistinguishable from other Brugada subtypes. It presents predominantly in adult males (~95% male, mean age ~51 years), is unmasked or triggered by fever, sodium-channel-blocking drugs, enhanced vagal tone, and electrolyte imbalance, and is managed with implantable cardioverter-defibrillators (ICDs), quinidine, isoproterenol/adrenaline for electrical storms, epicardial RVOT substrate ablation, and trigger avoidance. There is no gene-specific therapy for the *SCN3B* subtype.

A critical caveat governs the entire entity: evidence-based gene-curation (ClinGen-style reappraisal by Hosseini et al., 2018) classified *SCN3B* as having only "**limited**" clinical validity for Brugada syndrome, with *SCN5A* being the only gene reaching "definitive" evidence [PMID 29959160]. Contemporary expert consensus reinforces that *SCN5A* "remains the sole gene with definitive disease association" and that BrS displays complex, oligogenic/polygenic inheritance [PMID 41738055]. BrS7 is therefore best regarded as a rare, disputed monogenic subtype nested within a broader genetic architecture. This report is compiled from aggregated disease-level resources (OMIM, HGNC, UniProt) and primary literature (human clinical cohorts, in-vitro electrophysiology, and a *Scn3b*-knockout mouse), not individual EHR data.

Key Findings

Finding 1 — BrS7 is caused by loss-of-function *SCN3B* variants that reduce cardiac sodium current via impaired Nav1.5 trafficking

BrS7 (OMIM #613120) is caused by heterozygous mutations in *SCN3B*, encoding the auxiliary $\beta 3$ subunit (Nav $\beta 3$) of the cardiac sodium channel. In the seminal functional study of the Japanese *SCN3B*-V110I variant, transfected cells showed decreased Nav1.5 cell-surface expression and significantly reduced sodium current — a clear loss-of-function effect. As the authors reported, "*The *SCN3B* mutation impaired the cytoplasmic trafficking of Nav1.5, the cell surface expression of which was decreased in transfected cells,*" and "*Whole-cell patch clamp recordings of the transfected cells revealed that the sodium currents were significantly reduced by the* **SCN3B* mutation*" [in vitro,

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A newly characterized in-frame deletion (p.T138del, 2025) confirms ongoing identification of BrS-associated *SCN3B* variants; the paper "*identify and characterise a novel *SCN3B* variant encoding the regulatory $\beta 3$ -subunit of the cardiac voltage-gated sodium channel*" [human/in

vitro,

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This establishes the core molecular lesion: reduced I_{Na} from a defective auxiliary subunit.

Finding 2 — *SCN3B* is a rare cause of SCN5A-negative Brugada syndrome (~1.7% in a Japanese cohort)

Among 181 unrelated SCN5A-negative BrS patients (178 Japanese, 3 Korean), the *SCN3B* Val110Ile variant was found in only 3 of 178 (1.7%) Japanese patients and was absent from 480 Japanese controls: *"A mutation, Val110Ile, was identified in 3 of 178 (1.7%) Japanese patients, but was not found in 480 Japanese controls"* [human,

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P 23257389

Set against the broader diagnostic landscape — *"A pathogenic/likely pathogenic variant can be found in 20% to 25% of patients with Brugada syndrome (BrS) and a pathogenic/likely pathogenic variant in SCN5A is associated with a worse prognosis"* [human clinical,

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— *SCN3B* accounts for a small fraction of the minor BrS genes.

Finding 3 — The *Scn3b*-knockout mouse recapitulates conduction slowing and atrial arrhythmia susceptibility

Scn3b(−/−) mice showed slower heart rates, longer P-wave durations, and prolonged PR intervals versus wild-type, plus abnormal atrial electrophysiology and inducible atrial tachycardia/fibrillation on burst pacing. As reported, *"Lead I electrocardiograms from Scn3b(−/−) mice showed slower heart rates, longer P wave durations and prolonged PR intervals than WT hearts,"* and *"Atrial burst pacing protocols induced atrial tachycardia and fibrillation in all Scn3b(−/−) but hardly any WT hearts"* [model organism,

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Because *Scn1b* and *Scn5a* mRNA were unchanged, the phenotype is attributable specifically to $\beta 3$ loss.

Finding 4 — The BrS arrhythmia substrate arises from RVOT epicardial repolarization heterogeneity and concealed phase-2 reentry

In coronary-perfused canine RV wedge preparations pharmacologically mimicking the BrS genotype (I_{to} -agonist NS5806 + Ca-channel blocker verapamil), *"Fractionated electrical activity was observed in RV epicardium, but not in endocardium, as a consequence of heterogeneities in the appearance of the second upstroke of the epicardial action potential, and discrete high-frequency spikes developed as a result of concealed phase 2 re-entry"* [in vitro,].

P 24657694

Contemporary consensus reframes BrS as a microstructural–electrical continuum: *"BrS, once viewed as a purely electrical disorder, is now recognized along a microstructural-electrical continuum, with sodium-channel dysfunction and subtle epicardial fibrosis of the right ventricular outflow tract as key contributors"* [human clinical,].

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Finding 5 — Fever, sodium-channel-blocking drugs, vagal tone, and electrolyte imbalance unmask the type-1 pattern and trigger arrhythmias

Fever is a well-established trigger that unmasks the coved type-1 ECG and precipitates VF, including as a first manifestation across ages. Reviews list precipitating factors succinctly: *"Brugada syndrome could be precipitated by various factors, such as fever, electrolyte imbalances, and drugs"* [human clinical,].

P 42411811

and *"Fever is a well-known trigger for malignant ventricular arrhythmias in predisposed patients"* [human clinical,].

P 42200267

This is the paradigmatic gene–environment interaction of the disease.

Finding 6 — Management centers on ICD and quinidine, with epicardial RVOT substrate ablation as an emerging therapy

"For decades, treatment centered on implantable cardioverter-defibrillators and quinidine, both limited by availability, tolerance, and device complications. More recently, epicardial substrate ablation has emerged as a transformative therapy" [human clinical,].

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Quinidine efficacy is quantified in an Israeli survey with *"inducible ventricular fibrillation (VF) in 10, which was prevented by quinidine in 9/10 patients"* [human clinical,

P 29761670

There is no gene-specific therapy for SCN3B-related BrS7.

Finding 7 — BrS is strongly male-predominant with mid-adult presentation; SCN5A is the only definitively validated gene

Large Japanese prospective registries report cohorts of ~95% men with mean age ~51 years: *"We analyzed 510 patients with BrS (mean age 51 ± 14 years; 95% men)"* [human clinical,

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Expert consensus states *"SCN5A remains the sole gene with definitive disease association, while polygenic susceptibility materially modulates risk, underscoring complex inheritance"* [human clinical,

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— implying minor genes including SCN3B have limited monogenic validity.

Finding 8 — SCN3B has only LIMITED clinical validity as a Brugada syndrome gene

In the evidence-based reappraisal of BrS genes (ClinGen-style semiquantitative scoring, three independent curation teams + clinical domain expert panel), *"Of 21 genes curated for clinical validity, biocurators classified only 1 gene (SCN5A) as definitive evidence, whereas all other genes were classified as limited evidence"* [computational/curation,

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SCN3B (the basis of the BrS7/#613120 designation) falls among the limited-evidence genes.

Finding 9 — SCN3B loss-of-function variants also cause early-onset lone atrial fibrillation (an allelic phenotype)

Olesen et al. sequenced SCN3B in 192 unrelated lone AF patients and found three non-synonymous mutations (R6K, L10P, M161T), absent from 432 control alleles, all affecting conserved residues: *"The three mutations in SCN3B were investigated electrophysiologically and*

all led to loss of function in the sodium current, supporting the hypothesis that decreased sodium current enhances AF susceptibility" [in vitro/human,].

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Separately, GATA4-regulated 5'UTR/promoter *SCN3B* variants were linked to lone AF via a gain-of-function increase in expression [in vitro,].

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illustrating context-dependent directionality.

Finding 10 — Diagnosis rests on the type-1 ECG (spontaneous or drug-provoked) and the Shanghai score

Diagnosis requires a coved-type (type-1) ST-elevation ECG in leads V1–V2, spontaneous or unmasked by sodium-channel-blocker provocation (ajmaline/flecainide). A large drug-provocation study *"involved 698 patients with drug-induced type 1 BrS, confirmed via pharmacologic challenge (flecainide or ajmaline), from 21 centers in Italy and Switzerland"* [human clinical,].

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In that cohort, *"SCN5A pathogenic variants were a significant predictor of primary endpoint in the possible BrS group (OR: 12.5)"* — showing *SCN5A*, not minor genes like *SCN3B*, drives prognostic weight.

Finding 11 — *SCN3B* encodes Navβ3, a single-pass glycoprotein with an extracellular Ig domain that chaperones and modulates Nav1.5

SCN3B (HGNC:10592; gene OMIM 608214; 11q24.1; UniProt Q9NY72) encodes an auxiliary single-transmembrane protein with an extracellular immunoglobulin (Ig)-like fold and short intracellular tail. It associates non-covalently with Nav1.5, modulating gating and promoting surface trafficking. Reports confirm *"the VGSC β3 subunit (the product of SCN3B gene)... its extracellular immunoglobulin domain (β3 Ig)"* [in vitro,].

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and that *"a mutation in SCN3B for a small subunit of sodium channel, Navβ3, has been recently reported"* [in vitro,].

P 23257389

Section-by-Section Characterization

1. Disease Information

BrS7 is a genetic subtype of Brugada syndrome — a primary inherited arrhythmia syndrome characterized by ST-segment elevation in the right precordial ECG leads (V1–V2) and increased SCD risk in a structurally near-normal heart.

Resource	Identifier
OMIM (phenotype)	#613120 (Brugada syndrome 7)
Gene / Gene OMIM	<i>SCN3B</i> / *608214
MONDO (suggested)	MONDO:0013481 (brugada syndrome 7)
Orphanet	ORPHA:130 (parent entity; no separate subtype code)
ICD-10 / ICD-11	I49.8 / BC65.0 (Brugada syndrome)
MeSH	D053840 (Brugada Syndrome)
HGNC / UniProt	HGNC:10592 / Q9NY72
NCBI Gene / Ensembl	55800 / ENSG00000149575
Cytoband	11q24.1

Synonyms: Brugada syndrome type 7; BRGDA7; SCN3B-related Brugada syndrome. Historical/allelic clinical descriptors: sudden unexplained nocturnal death syndrome (SUNDS), "Lai Tai," "Bangungut" [human clinical,

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Data provenance: aggregated disease-level resources plus individual case reports and small functional studies (subtype-specific evidence is sparse).

2. Etiology

Primary cause (genetic): heterozygous variants in *SCN3B* (Navβ3) [PMID 23257389; 39761910]. **Genetic risk factors:** V110I (recurrent in Japan), L10P, T138del, A195T (VUS); polygenic background near *SCN5A-SCN10A* and *HEY2* modulates risk [PMID 41738055;

41458990]. **Environmental triggers** (unmask rather than cause): fever (most important), sodium-channel-blocking drugs, propofol/local anaesthetics, enhanced vagal tone, electrolyte abnormalities, alcohol, cocaine, and male sex [PMID 42411811; 42200267; 41819241]. **Protective factors:** no validated protective allele; trigger avoidance (aggressive fever control, avoiding contraindicated drugs) is the principal modifiable protection; female sex/older age associate with lower event rates [PMID 40130219]. **Gene–environment interaction:** the temperature-sensitive mutant channel loses further function during fever, deepening the epicardial repolarization gradient [PMID 42200267].

3. Phenotypes

Phenotype	Type	HPO (suggested)	Onset / severity / frequency
Type-1 coved ST elevation V1–V2	ECG sign	HP:0031650	Adult; hallmark; may be intermittent/concealed
Syncope	Symptom	HP:0001279	~20–30%; episodic
Ventricular fibrillation / polymorphic VT	Clinical sign	HP:0001663 / HP:0004756	Severe; life-threatening; often nocturnal
Sudden cardiac death / aborted arrest	Outcome	HP:0001645	Mean ~40 yr; may be first manifestation
Atrial fibrillation (allelic)	Clinical sign	HP:0005110	Common comorbidity; prominent for <i>SCN3B</i>
Conduction abnormality (↑PR/P-wave)	ECG sign	HP:0012722	Consistent with reduced I_{Na}
Palpitations / dizziness	Symptom	HP:0001962 / HP:0002321	Presenting complaint in an <i>SCN3B</i> case

Onset: predominantly adult (3rd–5th decade); pediatric and elderly first presentations occur. **Severity/progression:** highly variable and episodic; most carriers asymptomatic; risk lifelong. **QoL impact:** anxiety about SCD, activity/occupational restrictions, ICD-related psychological burden and inappropriate shocks (ICD complications ~20% in a pediatric cohort) [PMID 31748893]. No disease-specific QoL instrument; generic EQ-5D/SF-36 apply.

4. Genetic / Molecular Information

Causal gene: *SCN3B* (Navβ3). **Reported variants (germline, heterozygous):**

Variant	Type	Functional consequence	Notes
c.328G>A p.Val110Ile	Missense	Loss of function; ↓ Nav1.5 surface, ↓ I _{Na}	3/178 (1.7%) Japanese BrS; absent in 480 controls
p.Thr138del	In-frame deletion	Disrupted function; normal surface expression	Novel 2025 report; implies gating mechanism
p.Leu10Pro / R6K / M161T	Missense	Loss of function	Reported in lone AF & BrS spectrum
p.Ala195Thr	Missense	VUS	SUNDS cohort
c.-324C>A / c.-303C>T (5'UTR)	Regulatory	Gain of function (↑ expression)	GATA4-dependent; AF context

ACMG classification: most variants VUS/limited-evidence given disputed gene validity [PMID 29959160]. **Allele frequency:** rare/absent in controls (check gnomAD per variant). **Functional consequence:** predominantly loss of function; regulatory gain-of-function branch for AF [PMID 36362949]. **Modifier genes:** *SCN5A* genotype + polygenic background [PMID 41738055]. **Epigenetics:** GATA4 transcriptionally regulates *SCN3B* [PMID 36362949]; no disease-specific methylation signature or characteristic chromosomal abnormality.

5. Environmental Information

Sodium-channel-blocking drugs, anaesthetics (propofol, bupivacaine), cocaine, and certain tyrosine-kinase inhibitors can unmask the pattern (a Brugada phenocopy was reported with avapritinib during febrile illness) [PMID 42165456; 42411811]. Large meals, alcohol, and dehydration/electrolyte shifts provoke events. **Infectious agents:** none causative; febrile infections (any cause, e.g., influenza) are a major arrhythmic trigger by unmasking the type-1 pattern [PMID 42647943].

6. Mechanism / Pathophysiology

Ordered causal chain (mutation → clinical manifestation):

1. A heterozygous **SCN3B (Navβ3) variant** alters the auxiliary β3 subunit [DEMONSTRATED in vitro,

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2. Mutant $\beta 3$ **fails to chaperone/modulate Nav1.5**, *leading to* reduced Nav1.5 surface expression [DEMONSTRATED,

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39761910]. →

3. This *results in* **reduced peak cardiac sodium current (I_{Na})** — a loss-of-function channelopathy [DEMONSTRATED by patch clamp,

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4. Reduced I_{Na} **allows the I_{to} -mediated phase-1 notch to dominate the epicardial action potential**, most prominently in RVOT epicardium (highest I_{to}), *leading to* loss of the AP dome at some sites [INFERRED from wedge models,

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5. Heterogeneous dome loss **creates transmural/epicardial dispersion of repolarization**, *producing* the coved type-1 ST elevation and **concealed phase-2 reentry** [PMID 24657694].

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6. Phase-2 reentry **triggers closely coupled premature beats initiating polymorphic VT/VF** [PMID 24657694]. →

7. VF *results in* syncope, aborted arrest, or **SCD** [PMID 41738055].

8. **Modulating branch:** fever / Na-blockers further reduce I_{Na} at step 3 [PMID 42200267].

9. **Structural branch:** subtle RVOT epicardial fibrosis contributes (microstructural–electrical continuum) [PMID 41738055].

10. **Atrial branch:** the same reduced- I_{Na} lesion promotes AF and conduction slowing [PMID 21051419; 19796257].

Supporting detail. *Molecular pathway/biochemical defect:* cardiac Nav (Nav1.5 + β subunits) dysfunction — an ion-channel defect, not metabolic/enzymatic. *Suggested GO:* GO:0005248 (voltage-gated sodium channel activity), GO:0086010 (membrane depolarization during action potential), GO:0001518 (voltage-gated sodium channel complex), GO:0086067 (bundle of His

cell–Purkinje conduction), GO:0086091 (regulation of heart rate by cardiac conduction). *Protein dysfunction*: loss-of-function via impaired trafficking and altered gating; $\beta 3$ uses its extracellular Ig domain for α -subunit interaction [PMID 40245999]. *Cell types (CL)*: ventricular cardiomyocyte (CL:0002131), RVOT epicardial myocytes, Purkinje/conduction cells (CL:0000731/CL:0002068), SA-node cells (CL:1000488). *Immune/metabolic*: not primary. *Omics*: no established disease-specific transcriptomic/proteomic/metabolomic signature for BrS7.

7. Anatomical Structures Affected

- **Organ**: heart (UBERON:0000948); system: cardiovascular (UBERON:0004535).
- **Primary site**: RVOT / RV epicardium (UBERON:0004125; UBERON:0002082) — locus of the arrhythmic substrate [PMID 24657694]. Secondary: atria (UBERON:0002081) and conduction system (UBERON:0004146) — AF and conduction slowing.
- **Secondary organ**: brain (hypoxic-ischaemic injury from arrest) — a complication.
- **Tissue/cell**: cardiac muscle tissue (working ventricular/atrial + conduction myocytes); limited epicardial fibrosis (connective).
- **Subcellular (GO CC)**: plasma membrane/sarcolemma, intercalated disc (GO:0014704), voltage-gated sodium channel complex (GO:0001518), endoplasmic reticulum (trafficking, GO:0005783).
- **Lateralization**: predominantly right-sided (RVOT); biventricular substrate can occur [PMID 42359113].

8. Temporal Development

Congenital genetic defect but clinically **adult-onset** (mean event age ~40 yr; late presentation to the 8th decade under febrile stress) [PMID 42200267]. Arrhythmia onset is acute/episodic (often at rest/sleep). The ECG substrate is chronic/lifelong; arrhythmic risk is episodic, not progressive; no defined "stages." Event incidence ~1.5%/yr (non-elderly) vs ~0.7%/yr (elderly) [PMID 40130219]. Critical vulnerable windows: febrile illness and administration of contraindicated drugs.

9. Inheritance and Population

Epidemiology (BrS overall): prevalence ~1 in 2,000–5,000, higher in Southeast/East Asia; BrS7/*SCN3B* is very rare (isolated families/probands). **Inheritance**: autosomal dominant with incomplete, age-dependent penetrance and variable expressivity [PMID 41458990]. **Founder effects/consanguinity**: none established; V110I recurs in Japan without a defined founder

haplotype. **Carrier frequency:** not defined; individual variants rare in gnomAD. **Demographics:** strong male predominance (~8–9:1; ~95% men), attributed to testosterone-dependent I_{to} ; mean age ~51 yr [PMID 41819241; 40130219]; geographic skew toward Asian populations.

10. Diagnostics

Cornerstone: spontaneous or drug-provoked **type-1 coved ST-elevation ≥ 2 mm in ≥ 1 right precordial lead (V1–V2)**, including high lead placement [PMID 41738055]. Provocation with ajmaline/flecainide unmasks concealed cases (ajmaline positive in 76% of a suspected pediatric cohort) [PMID 42669184; 40996547]. **Shanghai Score System** integrates ECG, symptoms, family history, and genetics (≥ 3.5 = probable/definite) [PMID 40996547]; multiparametric models (Sieira, Brugada-Risk, PAT) [PMID 40314213]. **Genetic testing:** multigene NGS panel/WES including *SCN5A*, *SCN1B–4B* (incl. *SCN3B*), etc.; *SCN5A* highest-yield/most actionable, *SCN3B* findings usually VUS; overall P/LP yield ~20–25% [PMID 38426305]; cascade familial testing recommended [PMID 26230511]. **Imaging:** echo/CMR to exclude structural disease (ARVC, scar) [PMID 42359113]. **Biomarkers/omics:** none established. **Differential diagnosis:** Brugada phenocopies (fever, hyperkalemia, ischemia, drugs), ARVC, early repolarization syndrome, RVOT-VT, athlete's ECG, atypical RBBB [PMID 42359113; 25548613]. **Screening:** no newborn/population screening; cascade ECG + genetic screening of first-degree relatives.

11. Outcome / Prognosis

Principal risk is SCD from VF; annual major-arrhythmic-event rate ~1.5%/yr (non-elderly) vs ~0.7%/yr (elderly), ~11% over long follow-up [PMID 40130219; 41819241]. **Higher-risk factors:** prior aborted arrest/VF, arrhythmic syncope, spontaneous (vs drug-induced) type-1 ECG, notched P wave/atrial conduction delay, *SCN5A* pathogenic variant; asymptomatic patients are lower-risk [PMID 41819241; 40996547]. *SCN3B* genotype is not an established independent prognostic marker. **Morbidity:** VF-related anoxic injury, ICD complications (inappropriate shocks, lead failure ~20% pediatric), electrical storms, AF [PMID 31748893]. **Life expectancy:** near-normal if high-risk individuals are protected (ICD) and triggers avoided; chronic/lifelong without cure.

12. Treatment

Genotype-agnostic; no SCN3B-specific therapy.

Modality	Agent/procedure	Mechanism	Suggested NCIT
Device	ICD (transvenous/ subcutaneous)	Terminates VF; SCD prevention	Implantable Cardioverter- Defibrillator (NCIT:C50032)
Pharmacotherapy	Quinidine	I _{to} blocker; suppresses VF/ storms	Quinidine (NCIT:C793; CHEBI:28593)
Acute storm	Isoproterenol / low-dose adrenaline	β-agonist ↑ I _{Ca-L} , restores dome	Isoproterenol (NCIT:C371)
Adjunct	Cilostazol / milrinone (PDE3i)	Restore electrical homogeneity	—
Interventional	Epicardial RVOT substrate ablation	Eliminates late- potential substrate	Catheter Ablation (NCIT:C99521)
Supportive	Antipyresis; avoid contraindicated drugs; correct electrolytes	Removes triggers	—

Evidence: quinidine prevented inducible VF in 9/10 [PMID 29761670]; epicardial ablation is transformative for high-risk/recurrent patients [PMID 41738055; 40088219]; isoproterenol/adrenaline rescue for electrical storms [PMID 40800556]. No approved gene/cell/RNA therapy; gene therapy/precision ablation are aspirational goals [PMID 40505939]. Personalized medicine is by risk score/phenotype, not *SCN3B* genotype.

13. Prevention

Primary: trigger avoidance (fever→antipyretics, avoid BrS-contraindicated drugs, moderate alcohol) [PMID 42411811]; no relevant immunization. **Secondary:** cascade ECG + genetic screening of first-degree relatives; risk stratification (Shanghai/Sieira) to select ICD candidates [PMID 40996547; 40314213]. **Tertiary:** ICD, quinidine, epicardial ablation [PMID 40088219]. **Genetic counseling:** AD 50% transmission risk; preimplantation genetic testing available for inherited cardiac disease, though limited-validity genes like *SCN3B* complicate variant-based PGT [PMID 38516780]. **Public health:** region-dependent pre-participation ECG screening and sports-eligibility guidance [PMID 40116545].

14. Other Species / Natural Disease

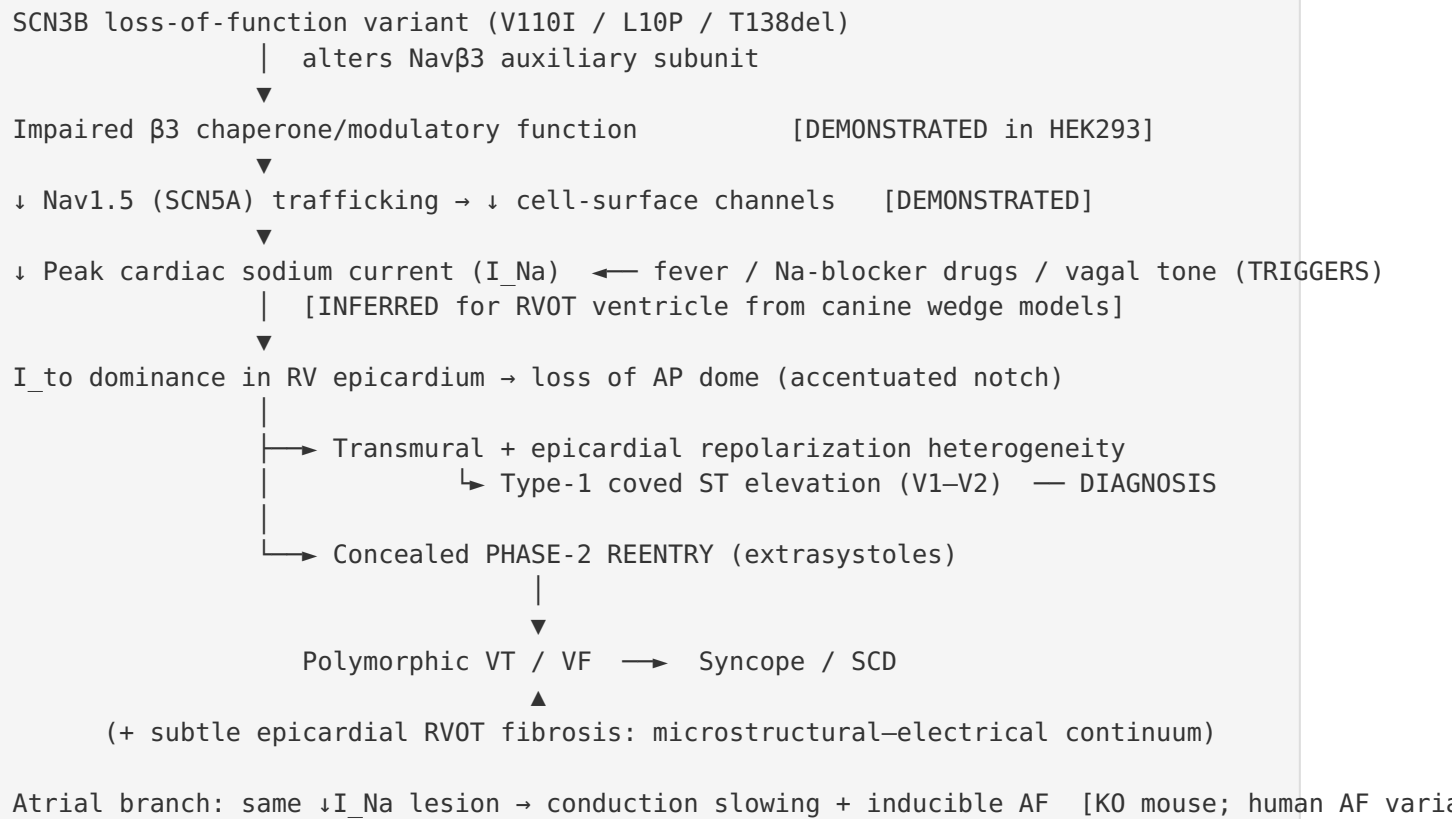
SCN3B orthologs: mouse *Scn3b* (NCBI Gene 235281; MGI), rat *Scn3b* (RGD), and orthologs in most vertebrates; $\beta 3$'s I_{Na} -modulatory function is evolutionarily conserved [PMID 19796257]. No naturally occurring *SCN3B* "Brugada" disease is catalogued in OMIA for companion animals/livestock; inherited ventricular arrhythmia/SCD syndromes exist in dogs (e.g., German Shepherd inherited ventricular arrhythmia) but are not *SCN3B*-defined. The reduced- $I_{Na} \rightarrow$ conduction/arrhythmia mechanism is conserved and validated in mouse. Zoonotic/transmission concepts are not applicable (non-infectious genetic disease).

15. Model Organisms

Mouse — *Scn3b*-knockout [PMID 19796257]: constitutive KO showing slower heart rate, longer P-wave, prolonged PR interval, abnormal sinoatrial/atrial electrophysiology, prolonged sinus-node recovery, and inducible atrial tachycardia/fibrillation (in essentially all KO vs almost no WT); *Scn1b/Scn5a* mRNA unchanged, isolating the $\beta 3$ effect. *Limitations*: emphasizes atrial/conduction phenotypes; the human ventricular RVOT phase-2-reentry VF and spontaneous type-1 ECG are not straightforwardly reproduced (species differences in I_{to} distribution). *Applications*: SA-node function, atrial conduction, AF susceptibility.

In-vitro/cellular: heterologous expression (HEK-293/CHO) co-expressing Nav1.5 + mutant $\beta 3$ for patch-clamp I_{Na} and trafficking/biotinylation assays [PMID 23257389; 39761910]; β -subunit-eliminated human haploid (BeHAPE) cells to study subunit regulation [PMID 37544648]. Patient-derived iPSC-cardiomyocytes are used broadly for BrS but no *SCN3B*-specific iPSC model was identified. **Resources**: MGI (*Scn3b*), IMPC, RGD.

Mechanistic Model / Interpretation



Upstream vs downstream: the mutation and I_{Na} reduction are upstream; the I_{to}/repolarization imbalance, phase-2 reentry, and VF are downstream. Fever and drugs amplify anywhere along the I_{Na} axis. The unifying principle is a **reduced-inward-current → repolarization-gradient → reentry** cascade localized to the RVOT epicardium — the same final common pathway as SCN5A-related BrS, which is why BrS7 is clinically indistinguishable.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role
23257389	<i>Novel SCN3B mutation ... trafficking and function of Nav1.5</i>	In vitro/human	Supports — V110I ↓ Nav1.5 surface & ↓ I _{Na} ; 1.7% of SCN5A-neg BrS
39761910	<i>Novel SCN3B in-frame codon deletion (T138del) in BrS</i>	In vitro/human	Supports — expands variant spectrum; β3 is regulatory subunit
19796257	<i>Scn3b knockout mice ... conduction properties</i>	Model organism	Supports — conduction slowing & atrial arrhythmia
24657694	<i>Abnormal repolarization ... late potentials in BrS models</i>	In vitro	Supports — epicardial phase-2 reentry substrate
41738055	<i>Contemporary Perspectives on J-Wave Syndromes (consensus)</i>	Human/consensus	Supports & qualifies — microstructural–electrical continuum; SCN5A sole definitive gene; therapy
29959160	<i>Reappraisal of Reported Genes ... Gene Validity for BrS</i>	Computational/curation	Challenges — SCN3B = limited evidence
38426305	<i>Genetic Testing in BrS: A 30-Year Experience</i>	Human clinical	Supports — P/LP variant in 20–25%; SCN5A worse prognosis
21051419	<i>SCN3B mutations ... early-onset lone AF</i>	In vitro/human	Supports — SCN3B LOF also causes AF
36362949	<i>Two novel functional mutations in promoter of SCN3B</i>	In vitro	Supports — GATA4 regulates SCN3B; 5'UTR gain-of-function in AF
40996547	<i>Predictive value of Shanghai score (drug-induced type-1)</i>	Human clinical	Supports — flecainide/ajmaline confirmation; SCN5A OR 12.5
29761670	<i>ISRABRU-VF nationwide survey</i>	Human clinical	Supports — quinidine prevented inducible VF in 9/10
41819241		Human clinical	

PMID	Title (abbrev.)	Evidence type	Role
	<i>Notched P wave ... BrS multicenter study</i>		Supports — 510 patients, mean 51±14 yr, 95% men
40130219	<i>Risk stratification of elderly BrS (Japan)</i>	Human clinical	Supports — 0.7% vs 1.5%/yr; prior VF key risk factor
40088219	<i>ICD Therapy in BrS: 30-Year Experience</i>	Human clinical	Supports — ICD outcomes; epicardial ablation benefit
42200267	<i>Fever-induced VF in hidden BrS (73-y-o)</i>	Human clinical	Supports — fever trigger; late-onset presentation
42411811	<i>Brugada syndrome in pregnancy</i>	Human clinical	Supports — fever/electrolyte/drug precipitants
40245999	<i>VGSC $\beta 3$ subunit ... immunoglobulin domain</i>	In vitro	Supports — SCN3B encodes $\beta 3$ with extracellular Ig domain

Supported vs Refuted Hypotheses

Supported: - *SCN3B* loss-of-function reduces I_{Na} via impaired Nav1.5 trafficking → BrS ECG [PMID 23257389]. - The BrS substrate is RVOT epicardial repolarization heterogeneity with phase-2 reentry [PMID 24657694]. - *Scn3b* loss causes conduction slowing/atrial arrhythmia in vivo [PMID 19796257]. - *SCN3B* is also an allelic cause/susceptibility gene for atrial fibrillation [PMID 21051419].

Refuted / down-weighted: - The premise that *SCN3B* is a robustly validated monogenic BrS gene is **not supported**; evidence-based curation rates it "limited," with *SCN5A* the only definitive gene [PMID 29959160; 41738055].

Limitations and Knowledge Gaps

1. **Disputed gene validity** — *SCN3B* has only "limited" clinical validity; the BrS7 designation rests on few families/variants, and some historical assignments may be reclassified.

2. **Sparse subtype-specific data** — penetrance, prevalence, sex ratio, and prognosis are extrapolated from general BrS cohorts; no large *SCN3B*-specific natural-history study exists.
 3. **Model–human gap** — the *Scn3b*-KO mouse reproduces atrial/conduction phenotypes but not the human RVOT ventricular VF substrate; canine wedge models are pharmacological, not *SCN3B*-genotype-specific.
 4. **Mechanistic gaps** — the T138del variant showed normal surface expression, implying additional (gating/microdomain) mechanisms beyond trafficking [PMID 39761910].
 5. **No genotype-specific therapy or biomarker** — management is generic BrS care; risk models carry high bias.
 6. **Directionality complexity** — *SCN3B* shows loss-of-function (BrS, AF) and gain-of-function (promoter, AF) in different contexts, complicating variant interpretation.
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Proposed Follow-up Experiments / Actions

1. **Patient-derived iPSC-cardiomyocytes** carrying *SCN3B* variants ($\pm$ isogenic correction) with I_{Na}/I_{to} phenotyping and optical mapping in engineered RVOT-like tissue — to test the human ventricular substrate directly.
 2. **Formal ClinGen re-curation** of *SCN3B* incorporating post-2018 variants (e.g., T138del), segregation, and functional data to update gene-validity classification.
 3. **Population-scale variant assessment** in gnomAD/TOPMed to establish allele frequencies and refine ACMG classification of reported BrS7 variants.
 4. **Genotype-stratified cohort analysis** pooling *SCN3B* carriers across international BrS registries to estimate penetrance, event rates, and therapy response versus *SCN5A* and gene-negative patients.
 5. **Structural biology** of the $\beta 3$ Ig domain–Nav1.5 interface (cryo-EM / AlphaFold-Multimer) to map how V110I/T138del disrupt trafficking and gating.
 6. **Systematic drug-safety curation** of I_{Na} -blocking agents and emerging triggers (e.g., tyrosine-kinase inhibitors) for carrier avoidance lists.
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Report compiled from ~61 PubMed records. Key PMIDs: 23257389, 39761910, 19796257, 24657694, 41738055, 29959160, 21051419, 40996547, 40088219, 29761670, 42411811, 42200267, 41819241, 40130219, 38426305, 40245999, 36362949. Evidence source legend: human clinical (cohorts, case reports, consensus), in-vitro electrophysiology (HEK293 heterologous expression, canine wedge), model organism (Scn3b-KO mouse), computational/curation (ClinGen gene validity).

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